



1. Description

1.1. Project

Project Name	RAK3172
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	01/16/2026

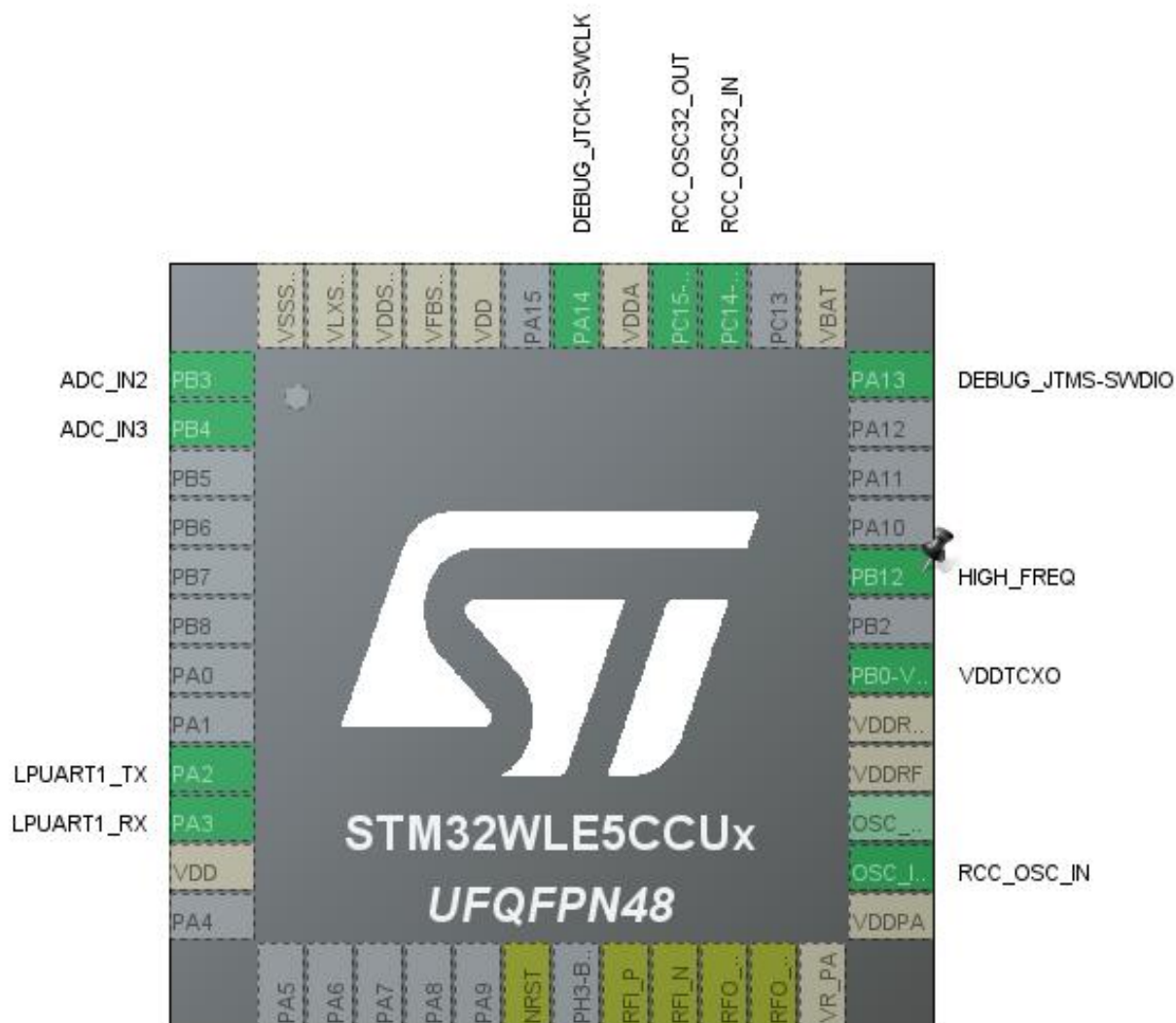
1.2. MCU

MCU Series	STM32WL
MCU Line	STM32WLEx
MCU name	STM32WLE5CCUx
MCU Package	UFQFPN48
MCU Pin number	48

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration

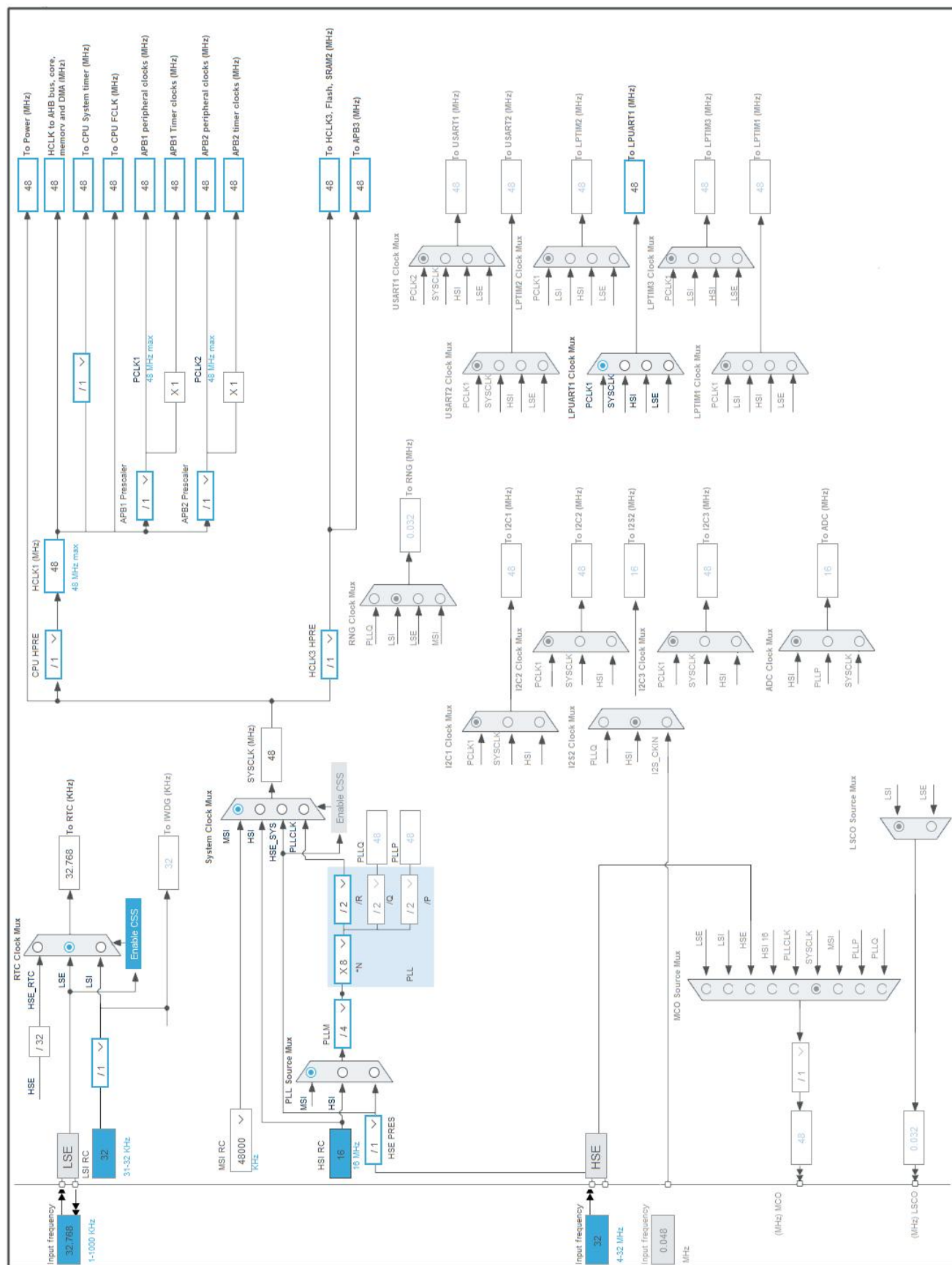


3. Pins Configuration

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PB3	I/O	ADC_IN2	
2	PB4	I/O	ADC_IN3	
9	PA2	I/O	LPUART1_TX	
10	PA3	I/O	LPUART1_RX	
11	VDD	Power		
18	NRST	Reset		
20	RFI_P	MonoIO		
21	RFI_N	MonoIO		
22	RFO_LP	MonoIO		
23	RFO_HP	MonoIO		
24	VR_PA	Power		
25	VDDPA	Power		
26	OSC_IN	MonoIO	RCC_OSC_IN	
28	VDDRF	Power		
29	VDDRF1V55	Power		
30	PB0-VDD_TCXO	I/O	VDDTCXO	
32	PB12 *	I/O	GPIO_Input	HIGH_FREQ
36	PA13	I/O	DEBUG_JTMS-SWDIO	
37	VBAT	Power		
39	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
40	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
41	VDDA	Power		
42	PA14	I/O	DEBUG_JTCK-SWCLK	
44	VDD	Power		
45	VFBSMPS	Power		
46	VDDSMPS	Power		
47	VLXSMPS	Power		
48	VSSSMPS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32WL
Line	STM32WLEx
MCU	STM32WLE5CCUx
Datasheet	DS13105 Rev7

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(AAA700)
Capacity	700.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	10.0 mA
Max Pulse Current	30.0 mA
Cells in series	1
Cells in parallel	1

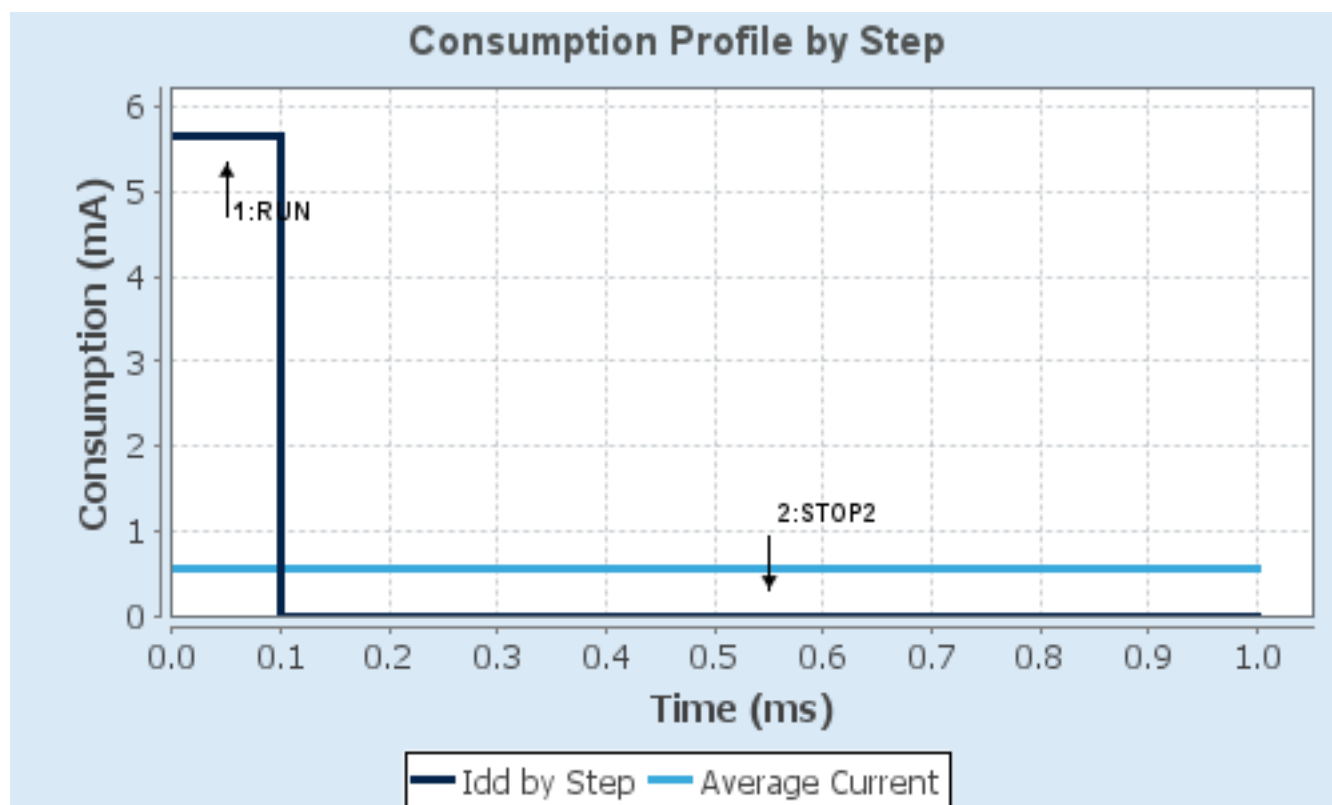
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Medium/SMPS-OFF	NoRange
Fetch Type	SRAM1	NA
CPU Frequency	48 MHz	0 Hz
Clock Configuration	MSI	ALL_CLOCKS_OFF
Clock Source Frequency	48 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	5.65 mA	885 nA
Duration	0.1 ms	0.9 ms
DMIPS	60.0	0.0
Ta Max	124.53	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	565.8 μ A
Battery Life	1 month, 21 days, 1 hour	Average DMIPS	60.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	RAK3172
Project Folder	C:\Users\danam\STM32CubeIDE\Alt-Workspace\RAK3172
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_WL V1.4.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC_Init	ADC
5	MX_LPUART1_UART_Init	LPUART1
6	MX_RTC_Init	RTC
7	MX_SUBGHZ_Init	SUBGHZ
8	MX_LoRaWAN_Init	LORAWAN

3. Peripherals and Middlewares Configuration

3.1. ADC

mode: IN2

mode: IN3

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Synchronous clock mode divided by 2
Resolution	ADC 12-bit resolution
Calibration	Disable
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Low Power Auto Wait	Disabled
Auto Off	Enabled *
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
DMA Continuous Requests	Disabled
Overrun behaviour	Overrun data overwritten *
Sequencer	Sequencer set to fully configurable
SamplingTime Common 1	160.5 Cycles *
SamplingTime Common 2	160.5 Cycles *
Oversampling Mode	Enabled *
Right Bit Shift	8 bit shift *
Triggered Mode	Single trigger
Trigger Frequency	High frequency
Ratio	Oversampling ratio 256x *

ADC_Regular_ConversionMode:

Enable Regular Conversions	Disable
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3.2. ADV_TRACE

mode: Enabled

3.3. DEBUG

JTAG and Trace: Serial Wire

3.4. LPUART1

Mode: Asynchronous

3.4.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.5. MISC

mode: Enabled

3.6. RCC

High Speed Clock (HSE): TCXO

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.6.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
MSI Calibration Value	0
MSI Auto Calibration	Enabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
LSE Drive Capability	LSE oscillator low drive capability

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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3.7. RTC

mode: Activate Clock Source

mode: Activate Calendar

Alarm A: Internal Alarm A

3.7.1. Parameter Settings:

General:

Asynchronous Predivider value
Bin Mode
SSRU Underflow Interrupt

RTC_PREDIV_A *

Free running Binary mode *

Enabled

Alarm A:

Free running 32 bit value
Binary AutoControl
Free running 32 bit mask

0

RTC_ALARMSUBSECONDBIN_AUTOCLR_NO *

SS[31:0] are compared and must match to activate alarm

3.8. SEQUENCER

mode: Enabled

3.9. SUBGHZ

mode: Activated

3.9.1. Parameter Settings:

Baudrate Prescaler Value **4 ***

3.10. SYS

Timebase Source: None

3.11. TIMER

mode: Enabled

3.12. TINY_LPM

mode: Enabled

3.13. LORAWAN

mode: Enabled

3.13.1. LoRaWAN application:

Application selection:

Application

Application configuration recommendations

End Node skeleton *

!! Please read carefully Information panel below!!

board settings:

Send Tx on Timer or Button Evt

Probes Lines in Platform Settings

TX_ON_TIMER

false

lora_app:

Active region

Transmission duty cycle

Application user port

Switch class port

Default class

Default handler message state

Handler Adaptive Data Rate

Default activation type

Force rejoin at each reboot

Default Tx output power

LORAMAC_REGION_US915 *

10000

2

3

CLASS_A

Unconfirmed message

On

OTAA

true

TX_POWER_0

Default Unicast ping slots periodicity 4
Default reponse timeout for class b and class c confirmed downlink frames in milli seconds. 8000

sys_conf:

Trace verbose level **VLEVEL_H ***
Enable Application Logging true
Disable Low Power Mode false
Enable Sensor false

3.13.2. LoRaWAN commissioning:

Commissioning:

Public network true
Current network ID 0

se-identity:

LoRaWAN device EUI 00,00,00,00,00,00,00,00
App/Join EUI 01,01,01,01,01,01,01,01
Application key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C
Network key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C
Device Address 00,00,00,00
Network session key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C
Application session key 2B,7E,15,16,28,AE,D2,A6,AB,F7,15,88,0
9,CF,4F,3C

lorawan_conf:

Enable Key read access true

3.13.3. LoRaWAN middleware:

lorawan_conf:

Region(s) selection please select the desired region(s) in the list below
Region Asia freq: 923 false
Region Australia freq: 915 false
Region China freq: 470 false
Region China freq: 779 false
Region Europe freq: 433 false
Region Europe freq: 868 true
Region Korea freq: 920 false
Region India freq: 865 false
Region USA freq: 915 true

Region Russia freq: 864	false
Enable Hybrid mode	false
Enable LoRaMAC ClassB	false
Enable the context management storage	true *
Select the LoRaWAN Link Layer specification version	v1.0.4
Enable the additional LoRaWAN packages	false
radio_conf:	
Radio maximum wakeup time (in ms)	1
radio_board_if:	
Select radio Driver	User Board
TCXO support	true
DCDC support	true
Tx Rfo Config	CONF_RFO_HP *
mw_log_conf:	
Enable Middleware log	true

3.13.4. Platform Settings:

ADC	ADC
RTC	RTC
USART	LPUART1

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC	PB3	ADC_IN2	Analog mode	No pull-up and no pull-down	n/a	
	PB4	ADC_IN3	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA13	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
LPUART1	PA2	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PB0-VDD_TCXO	VDDTCXO	n/a	n/a	n/a	
	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
GPIO	PB12	GPIO_Input	Input mode	Pull-up *	n/a	HIGH_FREQ

4.2. DMA configuration

DMA request	Stream	Direction	Priority
LPUART1_TX	DMA1_Channel1	Memory To Peripheral	Low

LPUART1_TX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	true	0	0
DMA1 Channel 1 Interrupt	true	0	0
LPUART1 Interrupt	true	0	0
RTC Alarms (A and B) Interrupt	true	0	0
SUBGHZ Radio Interrupt	true	0	0
PVD and PVM detector	unused		
FLASH (CFI) global Interrupt	unused		
RCC Interrupt	unused		
ADC Interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	false
RTC Tamper, RTC TimeStamp, LSECSS and RTC SSRU Interrupts	false	true	true
DMA1 Channel 1 Interrupt	false	true	true
LPUART1 Interrupt	false	true	true
RTC Alarms (A and B) Interrupt	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
SUBGHZ Radio Interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware									
LORAWAN									
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities
DMA	ADC	RTC	LPUART1				DEBUG		ADV_TRACE
GPIO			SUBGHZ						MISC
HWIC									SEQUENCER
RCC									TIMER
SYS									THRY_LPM

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32wl_bsdl.zip
HW Models	https://www.st.com/resource/en/hw_model/stm32wl_reference_board_mb1720-xo_ios_shield.7z
IBIS models	https://www.st.com/resource/en/ibis_model/stm32wl_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl_svd.zip
System View Description	https://www.st.com/resource/en/svd/stm32wl-svd.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1720_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_hp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1789_lp_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1791_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1792_bom.zip
Bill of Materials	https://www.st.com/resource/en/bill_of_materials/mb1842-bom.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1720_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_hp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_bdp.zip

Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1789_lp_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1791_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1792_manufacturing.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-bdp.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/mb1842-manufacturing.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1720_schematics.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_hp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1789_lp_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1791_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1792_schematic.zip
Schematic Pack	https://www.st.com/resource/en/schematic_pack/mb1842-schematic.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf

Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32cube-monitor-wireless-longrange_rfttest.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32wl5x-ex-product-line-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/beyond-the-wires-exploring-bluetooth-and-lorawan-connectivity.pdf
Flyers	https://www.st.com/resource/en/flyer/flnucleolrwan.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wl5ex.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-

timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5406-how-to-build-a-lora-application-with-stm32cubewl-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5409-stm32cube-mcu-package-examples-for-stm32wl-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5457-rf-matching-network-design-guide-for-stm32wl-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5480-how-to-build-a-sigfox-application-with-stm32cubewl-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5042-how-to-calibrate-the-hse-clock-for-rf-applications-on-stm32-wireless-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5407-how-to-optimize-the-rf-board-layout-for-stm32wl5xex-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5886-guidelines-for-design-and-board-assembly-of-land-grid-array-packages-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5646-how-to-select-a-

32-mhz-hse-oscillator-for-stm32wl5xex-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf

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features-of-stm32wl5xex-microcontrollers-stmicroelectronics.pdf

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